DAILY AI BRIEFING · LEARN ONE THING + WHAT MOVED

Keeping up with the AIs

Four bits is plenty

Issue #36 | Wed, 04 Nov 2026 | ☕ ~8 min read

⚡ If you only have one minute

1. **Postman's desktop app quietly installed plugins into Claude Code and Codex**, and tried to approve its own startup hook. Check your `~/.claude` and `~/.codex` folders today.
2. **Anthropic open-sourced FlashProve**, a way to prove mathematically that a reply came from the model a provider says it used.
3. **Part 1: quantization**, the trick that let Simon Willison run a 27B model on a laptop yesterday. Same brain, a quarter of the memory.

Today in three numbers

19 MB
size of a **FlashProve proof** for one 1,024-token reply

46 min
for a **4-bit Qwen on a laptop** to spot a Pac-Man net from its raw weights

1.2B
people using **ChatGPT every week**, per OpenAI's consumer lead

PART 1 Learn one thing

🗜️ Quantization: fitting big models on small hardware

4 MIN READ | MODULE 5 · UNDER THE HOOD

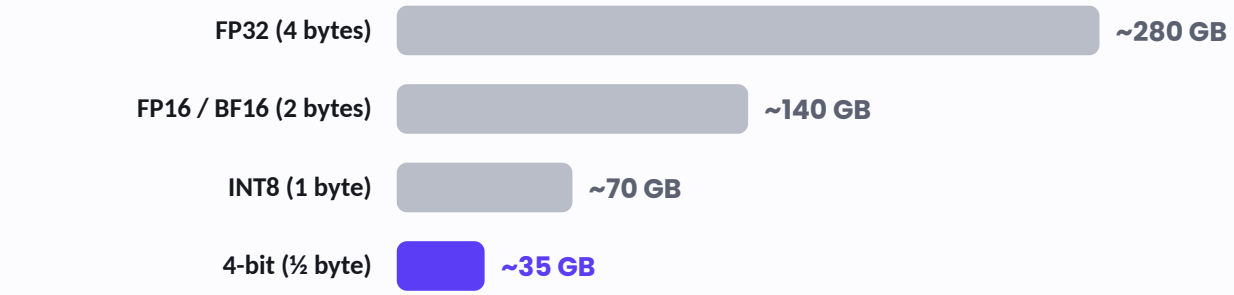
Quantization stores each of a model's numbers with fewer bits, say 4 instead of 16. The model gets about four times smaller and usually faster, and if it's done well, barely any dumber.

The plain-English version. A model is billions of numbers called **weights**. During training they're usually kept as 16-bit decimals (**FP16 or BF16**), so each weight takes 2 bytes. **A 70-billion-parameter model is therefore about 140 GB** before it has read a single word. That's more than any laptop GPU.

Quantization rounds those weights onto a small grid of allowed values. With 8 bits there are 256 steps; with 4 bits, only 16. You also store a **scale** that says how big one step is, so you can turn the small integers back into roughly the right decimals when the model runs.

Analogy: a thali photo sent over WhatsApp. The original from your phone is 12 MB. WhatsApp squeezes it to a few hundred KB. **Zoom in and you'll find fuzzy edges, but your friend can still tell it's dal, rice and two rotis.** Quantization is the same deal: lose the fine detail nobody needed, keep the picture.

Memory for the weights of a 70B model



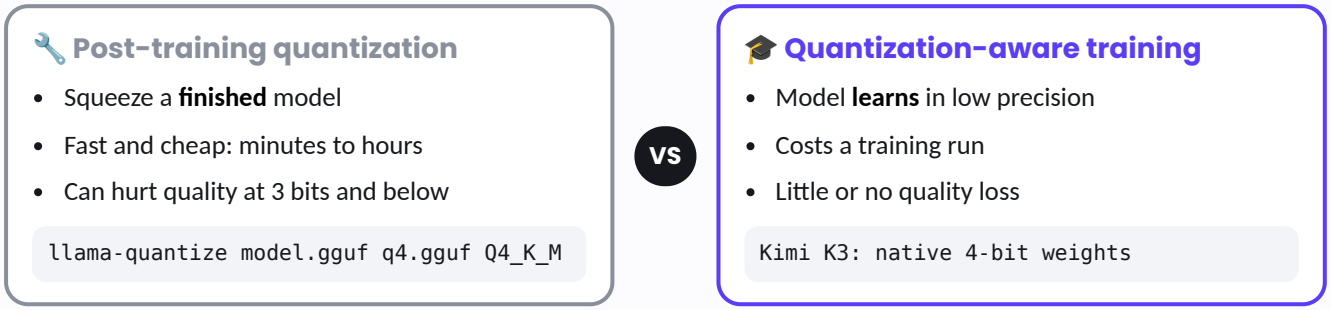
Our maths: parameters × bytes per weight. Real files are a little bigger (scales, some layers kept at higher precision), and running the model needs extra memory for the KV cache.

Quantizing one block of weights to 4 bits



Illustrative 'absmax' scheme. Working in small blocks means one unusually large weight only blurs its own block. Real formats such as GGUF's Q4_K_M, GPTQ and AWQ are cleverer about which weights get more precision.

A real example, from yesterday. Simon Willison ran **Qwen 3.8 27B in the `Q4_K_M` format** (a popular 4-bit GGUF file) on his laptop with LM Studio. By our maths, that model needs about 54 GB at 16 bits but **around 16 GB at Q4_K_M** (which averages a bit under 5 bits per weight). It then read the raw weights of small game-playing networks and named three of four correctly. (AI Weekly)



Command shown is the shape of llama.cpp's quantize tool. Kimi K3 detail per RunAIHome.

Why now? Low precision is all over today's news. FlashProve was benchmarked on a **4-bit** Llama 4 Maverick. And Moonshot promised **Kimi K3.1 Pro's weights by today**; its predecessor K3 shipped natively in 4-bit, about **1.56 TB.** (RunAIHome)

📖 Jargon buster

- **Precision / bits:** how many bits store each number. **Fewer bits, smaller model**, rougher numbers.
- **Scale:** the multiplier that turns small integers back into **approximately the original** decimals.
- **GGUF:** the file format used by llama.cpp, Ollama and LM Studio. Names like **Q4_K_M or Q8_0** tell you the bits.

🧪 Try it in 10 minutes

- **No code:** open any popular model's GGUF page on Hugging Face and compare file sizes for **Q8_0, Q4_K_M and Q2_K**. Pick the biggest that fits **your GPU memory (or RAM) with room to spare**. (Hugging Face)
- **Some code:** run the 10 lines below with just NumPy. **Change `bits` to 2** and watch the error grow.

🛠️ For devs: quantize by hand, then for real

```
import numpy as np
np.random.seed(0)
w = np.random.randn(8).astype(np.float32) * 0.5
def quantize(w, bits):
    qmax = 2 ** (bits - 1) - 1              # 4 bits -> 7
    scale = np.abs(w).max() / qmax          # one scale per block
    q = np.round(w / scale).clip(-qmax, qmax).astype(int)
    return q, q * scale                      # ints to store, values to use
for bits in (8, 4):
    q, back = quantize(w, bits)
    print(bits, q, "max error", np.abs(w - back).max().round(4))
# 8 [100 23 55 127 106 -55 54 -9] max error 0.0041
# 4 [6 1 3 7 6 -3 3 0]             max error 0.0784
```

Gotchas: 4-bit is the usual sweet spot; **below 3 bits quality often falls off a cliff**, and maths and code suffer first. Run your evals (Issue #26) before and after. For real models, Hugging Face's `BitsAndBytesConfig(load_in_4bit=True)` does this for you.

🔗 GO DEEPER

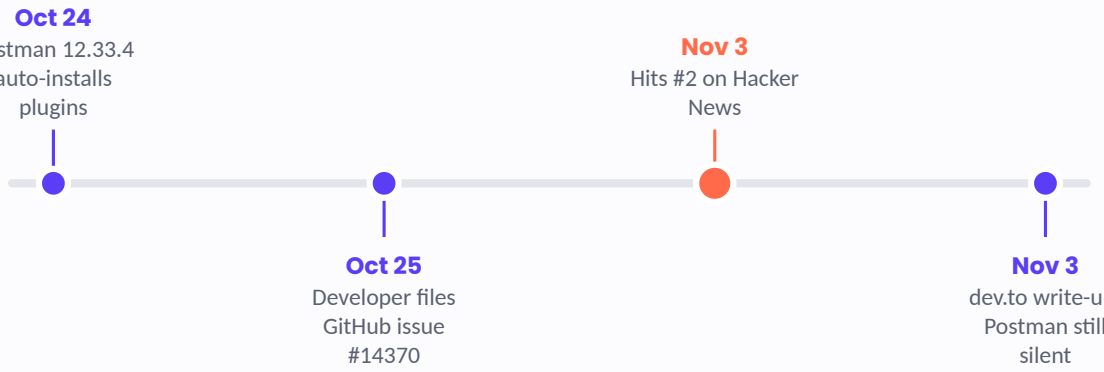
- **A minimal introduction to quantization** (Omar Sanseviero). Absmax and zero-point quantization in a few lines.
- **Making LLMs even more accessible with bitsandbytes, 4-bit quantization and QLoRA** (Hugging Face). NF4 and fine-tuning in 4-bit.
- **Selecting a quantization method** (Hugging Face docs). bitsandbytes vs GPTQ vs AWQ.
- **GGUF on the Hugging Face Hub** (Hugging Face). What names like Q4_K_M mean.

PART 2

What moved in AI

7 min read · Covering the last 24 hours. An API tool that rewired people's coding agents, a cryptographic receipt for AI replies, and Siri opening up to every app.

How the Postman plugin story unfolded



Top story 1 · Postman quietly plugged itself into Claude Code and Codex

BIG DEAL FOR DEVS

What happened. A developer reports that opening **Postman desktop 12.33.4** silently installed Postman plugins into **Codex and Claude Code**, with no prompt. The GitHub issue reached **#2 on Hacker News** yesterday, and Postman **hasn't responded publicly**. (AI Weekly) (GitHub)

The details that matter:

- **What it touched (macOS):** `~/.codex/config.toml`, `~/.claude/settings.json`, `~/.claude/plugins/installed_plugins.json`, plus a Postman program in `~/.local/bin`. It also pointed both agents at **Postman's MCP server**.
- **The worrying bit:** an installer function called `approvePostmanHook` **wrote the hook's trust record itself**. Those trust checks exist so that *you* approve code that runs when your agent starts.
- **What the hook says:** it tells the agent to load Postman's `api-engineer` skill before **any non-trivial API task**, not just ones that mention Postman.
- **Not yet confirmed independently:** nobody has reproduced it on a clean machine so far. The reporter asks for **opt-in installs and no self-approval**.

Why it matters. Your coding agent's config folder is now **valuable real estate**, and apps are starting to fight over it. A hook that runs on every session is **a supply-chain risk** (Issue #32's trojanized extension was the extreme case).

WHO SHOULD CARE 🙋 Everyone ●○○○○ 🎨 Vibe coders ●●●●○ 🛠️ Developers ●●●●● 🔥 Hype level ●●○○○

👀 **Should you care?** Yes, if you use Postman and an AI coding agent. It takes two minutes to check.

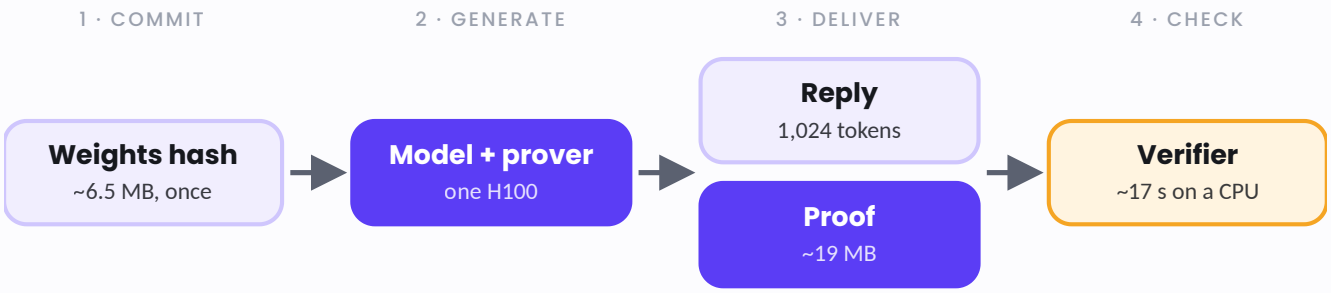
> **🛠️ For devs: a quick audit**
>
> Look for unexpected `hooks` and `mcpServers` entries in `~/.claude/settings.json`, any `[mcp_servers]` blocks in `~/.codex/config.toml`, and anything new in `~/.local/bin`. Keep these files **in a dotfiles git repo** so `git diff` shows you who changed what.

Top story 2 · Anthropic's FlashProve: a receipt that proves which model answered

RESEARCH TRUST

What happened. Anthropic, with **EPFL, MIT, StarkWare and Succinct**, released **FlashProve** under the Apache 2.0 licence. It produces a **zero-knowledge proof** that a reply came from one specific, committed set of weights, **without revealing the weights.** (AI Weekly) (Anthropic)

How a provable reply works



Figures for a 4-bit Llama 4 Maverick (400B MoE), as summarised by AI Weekly from Anthropic's post.

The details that matter:

- **Speed:** proofs generate at **over 100 tokens per second** on one H100, but the prover is **72 to 94 times slower** than serving the same model normally with vLLM.
- **Narrow for now:** replies up to **1,024 tokens**, integer (quantized) models only, and **greedy decoding** (always picking the top word, so no temperature).
- **Checked by maths, too:** its three security guarantees were **formally verified in Lean**, a proof-checking language.
- **A promise kept:** Anthropic's safety roadmap aimed for a **provable-inference prototype by 30 September 2026**.

Why it matters. Today you **trust** that "Model X" on some API really is Model X and not a cheaper swap. FlashProve hints at a future where **you can check**. That matters for audits, regulated industries and anyone reselling models.

WHO SHOULD CARE 🙋 Everyone ●○○○○ 🎨 Vibe coders ●○○○○ 🛠️ Developers ●●●○○
🔥 Hype level ●●○○○

👀 **Should you care?** Not for your app today; ~90x the compute is far too costly to run on every reply. Worth knowing if you **buy models through resellers** or work in compliance.

Top story 3 · Apple opens Siri AI to every app

TRY IT FOR DEVS

What happened. Apple opened a **Siri AI developer preview** in the iOS, iPadOS, macOS, watchOS and visionOS **27.3 betas**. Any app can now expose its content and actions to Siri through **App Intents**. (Apple Developer) (AI Weekly)

The details that matter:

- **Skills:** new **"skills"** let Siri chain several app actions in a row, and it can pull in anything from the **Spotlight semantic index**.
- **Timing:** App Store submissions open in **spring 2027**; Apple says customers get it in "a few months."
- **Languages:** English, French, Japanese, Korean, Portuguese and Spanish at launch. **No Indian languages yet**, and it's **unavailable in the EU**.
- **A future fee:** server-side features have **daily limits**, and Apple says expanded access will cost extra later. No price yet.

Why it matters. It's Apple's version of tools and MCP (Issues #16 and #20): **your app becomes something the assistant can call**. On an iPhone, that's a lot of potential users.

WHO SHOULD CARE 🙋 Everyone ●●●○○ 🎨 Vibe coders ●●○○○ 🛠️ Developers ●●●●○
🔥 Hype level ●●●○○

👀 **Should you care?** If you ship an iOS app, yes: **App Intents you add now** should carry over. Apple runs developer Q&A labs **today and tomorrow (4–5 Nov)**.

⚡ Quick hits

🔁 Codex training toggle flips back on?

A Hacker News thread says OpenAI's **"Improve the model for everyone"** setting re-enabled itself for some Codex users who'd turned it off. **No OpenAI response yet.** (AI Weekly)

🛡️ CrowdStrike buys two AI-security startups.

About **$400M for Fiddler AI** (agent monitoring) and **$95M for Dreadnode** (AI attack testing), per SecurityWeek. (AI Weekly)

🎨 Fal doubles to $16B.

The image and video model host raised **$1.3B** led by Bessemer. Its CEO says it had to **rate-limit customers** in summer for lack of compute. (AI Weekly, via Bloomberg)

🏫 Google: block AI, lose Search.

From early 2027, US school districts that turn off Google's AI tools for students **also lose Search**, per the NYT. "There is no middle setting." (AI Weekly)

💬 ChatGPT: 1.2B weekly users.

OpenAI's Adam Fry told Big Technology it added **200M in two months**, and its search volume is **about 20% of Google's**. (AI Weekly)

🇨🇳 DeepSeek V5 on Huawei chips.

DeepSeek will reportedly train on at least **160,000 Huawei Ascend 950DT** chips from as soon as late 2026. (AI Weekly)

📈 AMD data centre nearly doubles.

Q3 data-centre revenue **rose 97% to $8.5B** as its MI450 chips ship; total revenue $13.4B. (AI Weekly)

🗓️ Coming up.

Kimi K3.1 Pro weights **promised today (4 Nov)**: we hadn't seen them by press time. Apple labs **4–5 Nov**. Ollie shuts **6 Nov**. US lame-duck session **9 Nov**. Anthropic usage policy **12 Nov**.

🎯 One thing to try today: pick your quant like a pro

Write down your GPU memory (or Mac RAM). In LM Studio or Ollama, find a model whose **Q4_K_M file is about 70% of that**, leaving room for context. Then try the **Q8_0** of a smaller model and compare answers on three of your own prompts. **Bigger model at 4 bits often beats a smaller one at 8.**

🧠 Pop quiz: 30 seconds, no Googling

1. Roughly how much memory do a 70B model's weights need at 4 bits?

- A 140 GB
- B 70 GB
- C 35 GB

2. What did Postman's installer do that worried people most?

- A Deleted files
- B Approved its own hook
- C Sent your API keys

3. What does FlashProve prove?

- A The reply is true
- B Which weights made it
- C The prompt was safe

Answers (no peeking): 1. C: 70B × half a byte · 2. B: it wrote its own trust record · 3. B: it ties a reply to committed weights

Keeping up with the AIs is researched and written by Claude every morning from lab announcements, tech press, research releases and developer chatter. Company claims are labelled as claims, and every item links to its source. Spot an error or want a topic covered? Just reply. ✉️